Table 1: Clustering quality results for the CREDITCARD dataset.

					Lloyd		Cost					$\text{Gap}^{\text{SFC}}_{\text{Cost}}$ [%]				Balance					
k	λ	Target	p	q	Cost	Bal.	MIQCP	SetVars	MPFC	Flow	SFC	MIQCP	SetVars	MPFC	Flow	Lloyd	MIQCP	SetVars	MPFC	Flow	SFC
2	0.01	0.65	475	731	4,717	0.56	-	4,798	4,727	**4,726**	5,174	-	-7.27	-8.64	-8.65	✗	-	✓	✓	✓	✓
2	0.10	0.59	433	733			-	4,793	4,719	**4,718**	5,039	-	-4.90	-6.36	-6.37	✗	-	✓	✓	✓	✓
2	0.30	0.46	453	986			-	4,740	4,718	**4,717**	4,807	-	-1.39	-1.86	-1.86	✓	-	✓	✓	✓	✓
5	0.01	0.65	341	525	3,301	0.54	-	3,614	**3,313**	3,316	4,177	-	-13.47	-20.69	-20.61	✗	-	✓	✓	✓	✓
5	0.10	0.59	62	105			-	3,337	3,304	**3,303**	3,749	-	-11.00	-11.88	-11.90	✗	-	✓	✓	✓	✓
5	0.30	0.46	62	135			-	3,342	**3,302**	**3,302**	3,450	-	-3.14	-4.28	-4.29	✓	-	✓	✓	✓	✓
10	0.01	0.65	455	701	2,564	0.49	-	3,084	**2,603**	2,615	3,806	-	-18.99	-31.62	-31.30	✗	-	✓	✓	✓	✓
10	0.10	0.59	95	161			-	2,732	**2,574**	2,575	3,045	-	-10.27	-15.45	-15.42	✗	-	✓	✓	✓	✓
10	0.30	0.46	95	207			-	2,611	2,569	**2,567**	2,857	-	-8.59	-10.06	-10.14	✓	-	✓	✓	✓	✓
15	0.01	0.65	455	701	2,247	0.44	-	2,797	2,317	**2,311**	3,629	-	-22.92	-36.16	-36.33	✗	-	✓	✓	✓	✓
15	0.10	0.59	95	161			-	2,427	2,276	**2,268**	2,878	-	-15.65	-20.91	-21.21	✗	-	✓	✓	✓	✓
15	0.30	0.46	95	207			5,837	2,340	2,258	**2,253**	2,587	125.61	-9.55	-12.73	-12.93	✗	✓	✓	✓	✓	✓
Average												125.61	-10.60	-15.05	-15.08						

(-) no feasible solution; (**bold**) best result; (underline) 2nd best result; (✓) target balance attained; (✗) target balance not attained

Table 2: Running times for the CREDITCARD dataset.

					CPU_{tot} [s]					CPU_{avg} [s]			$\text{Gap}^{\text{SFC}}_{\text{CPU}_{\text{avg}}}$ [%]		Avg. iterations	
k	λ	Target	p	q	MIQCP	SetVars	MPFC	Flow	SFC	MPFC	Flow	SFC	MPFC	Flow	MPFC	Flow
2	0.01	0.65	475	731	limit	limit	1,916.70	177.37	limit	19.17	1.77	129.80	-85.23	-98.64	4.90	4.63
2	0.10	0.59	433	733	limit	limit	1,838.68	189.72	limit	18.39	1.90	82.31	-77.66	-97.69	5.00	5.02
2	0.30	0.46	453	986	limit	limit	1,806.72	187.24	2,519.71	18.07	1.87	25.20	-28.29	-92.58	4.98	4.95
5	0.01	0.65	341	525	limit	limit	limit	427.72	limit	102.86	4.28	96.31	6.80	-95.56	8.60	7.34
5	0.10	0.59	62	105	limit	limit	limit	445.42	1,915.45	75.14	4.45	19.15	292.38	-76.76	8.35	7.69
5	0.30	0.46	62	135	limit	limit	limit	457.87	1,223.76	66.72	4.58	12.24	445.10	-62.58	8.39	8.00
10	0.01	0.65	455	701	limit	limit	limit	698.34	limit	212.05	6.98	123.60	71.56	-94.35	8.88	7.45
10	0.10	0.59	95	161	limit	limit	limit	758.50	2,612.05	157.14	7.59	26.12	501.61	-70.94	7.83	8.20
10	0.30	0.46	95	207	limit	limit	limit	783.74	1,465.62	133.60	7.84	14.66	811.32	-46.52	8.48	8.61
15	0.01	0.65	455	701	limit	limit	limit	1,120.10	limit	401.27	11.20	123.86	223.97	-90.96	10.33	8.88
15	0.10	0.59	95	161	limit	limit	limit	1,139.02	2,628.90	276.93	11.39	26.29	953.37	-56.68	9.92	9.04
15	0.30	0.46	95	207	limit	limit	limit	1,195.27	1,475.65	257.85	11.95	14.76	1,646.95	-19.04	11.00	9.63
Average													396.82	-75.19	8.06	7.45

(limit) time limit of 3,600 seconds reached

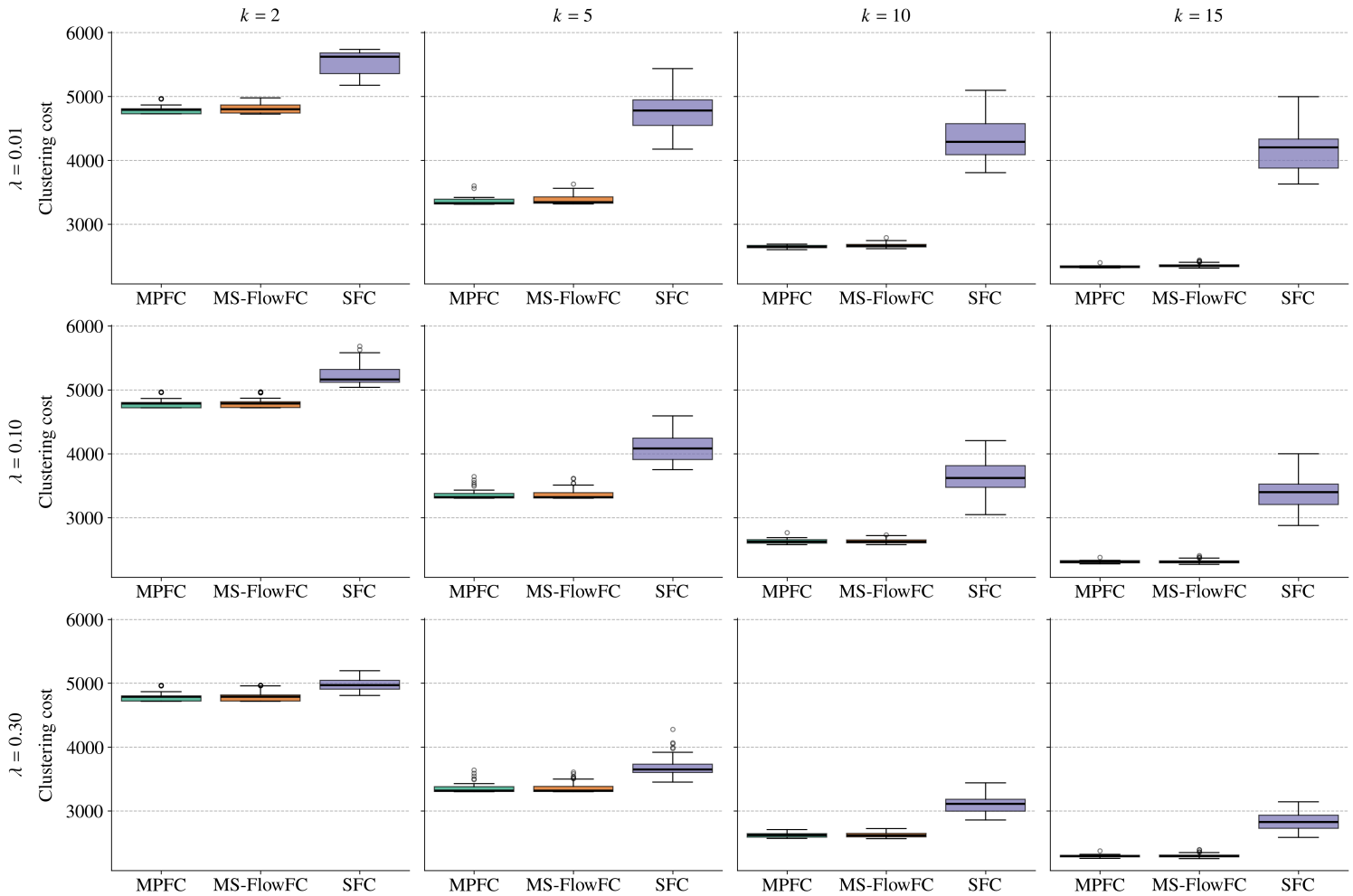

Figure 1: Boxplots of clustering costs for solutions obtained with different random seeds on the CREDITCARD dataset, across different methods, numbers of clusters, and tolerance values.